

# RESPONSE TO THE REFEREE: NEURAL BELLMAN OPERATORS

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Revision R52, 8 October 2026. This response addresses the R50 threshold report and the controlling R49 substantive report. The same Neural Bellman Operators paper is revised, with all historical adverse evidence retained.

## 1. RESPONSE TO THE REFEREE: NEURAL BELLMAN OPERATORS

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We thank the referee for separating the quality of the implemented economic policy from the sharpness of its numerical certificate, and for requiring one complete, reproducible submission. We address the latest R50 threshold report and its controlling R49 substantive report. We retain the original title, economic primitives, Bellman objective and broader theoretical development. This is a revision of *Neural Bellman Operators*, not a replacement paper with a different topic.

The threshold report is fixed at commit 2822f50100c7a53ec5fe07d39e9b37d487ab0547, and the substantive report at 4708450610c38e6b8963670cecad887508c85c0. The complete R51 predecessor is 62ffa8d0753c82a1363831f49a297c815d89204f. R51 had already integrated the previously dispersed source and evidence. R52 starts from that complete object rather than the input-only R50 branch. The earlier response is retained unchanged under preserved/R51/response.md.

### 1.1. 1. *Principal mathematical and empirical changes*

The new section, **Policy-relevant continuation errors**, strengthens the signed-width finite-sweep theorem. Let  $e$  denote the difference between the incumbent's true policy value and its fitted continuation. Rather than charge its global range width  $w$  at every action comparison, we certify

$$|(P_t^a - P_t^b)e_{t+1}^k(x)| \leq R_t^k(x; a, b) \leq \chi_t^k.$$

The proposal is accepted only when its centered comparison upper bound plus beta times the proposal–incumbent action-error certificate is nonpositive. Otherwise the incumbent is retained exactly. The theorem proves both all-state nonworsening and

$$E_t^{k+1} \leq \beta_t E_{t+1}^k + \alpha_t^k + \zeta_t^k + 2\beta_t \chi_t^k.$$

The original finite-horizon unrolling therefore holds with chi in place of w. The global-width rule remains a valid fallback. Exact action contrasts, search and comparison imply optimality after at most T passes even with nonzero, nonconstant global continuation errors. The coefficient two remains sharp; its proof is retained and specialized to the new certificate class. These statements concern the original Bellman optimum, not a weakened objective.

A coupled-residual proposition constructs pairwise error bounds using policy Bellman residuals and couplings with the correct marginals. It does not assume access to the true policy value or success of an unexecuted transport optimizer. The action-null corollary identifies precisely which nonconstant continuation errors can be removed from action comparisons. Membership must be proved, or any remaining contrast must be bounded; it is never inferred from sampled residuals.

In the original investment dynamics the first two action coefficients are one-half and one-quarter. Consequently the common-innovation difference of the first coordinate and twice the second is independent of the action. The nuisance  $M \setminus \text{ReLU}(y_1 - \sqrt{2} y_2 + 1)$  is therefore action-null, although its global width is  $2M$ . This is an explicit neural-realization robustness result. It is shared by a native representation of the same function and is not relabeled as neural superiority.

The additional experiment enumerates all 65,536 closed eight-bit observation cells for each of the eight retained target-two policies. Four amplitudes, zero, one, sixteen and 4096, were fixed before this execution. The study retains whole-cell feasibility, the original terminal action lattice and proposal, outward cost comparisons, and

every rejected proposal. It records all 524,288 policy–cell cases and every amplitude gate. Its returned contrast-aware actions are independently checked against the unmodified inherited terminal implementation. All new counts and clocks are generated from these records.

This is an exhaustive deterministic robustness ablation, not another independent expected-cost sample. The continuous investment study still executes a final-date improvement, not T full policy-iteration passes. The latter are exercised separately by exact finite-state regression programs. The manuscript does not interchange these two forms of evidence.

## 1.2. 2. R50 threshold requirements

**4.1: Canonical paper object.** The new revision directory contains ordinary main, supplementary, complete-development and response sources. The intended canonical branch is revision/econometrica-nbo-r52-review-ready-2026-10-08; it is created only after the numerical-record audit, tests, five-document compilation and clean archive rebuild have succeeded. Source and evidence remain permanently committed, not dependent on an expiring Actions artifact.

**4.2: Substantive response and exact changes.** The new theoretical section is sections/action\_contrast52.tex, its full proofs are sections/contrast\_proofs52.tex, and the new original-economy study is sections/study52.tex. Exact predecessor-to-current differences and label preservation are generated for all four manuscript editions. No R49/R51 adverse observation is replaced. This response covers every substantive blocking and major comment below.

**4.3: Numerical evidence.** We retain the full R51 construction, prospective resource allocation, genuine surplus comparator, common-accuracy block and 280 direct-cost estimands. R52 adds a distinct exhaustive certificate diagnostic and charges its own proposal, comparison, inherited-policy replay, serialization and memory costs. It does not splice historical construction clocks into a new service or pool identical policies into a larger sample.

**4.4: Publication integrity.** All new inputs and code are fixed before the new ablation. Every cell file is hash-bound and replayed independently of the reported

counts. The inherited evidence archives, source-freeze manifests and exact source files are rechecked. The final delivery record binds candidate Git objects, sources, evidence and compiled outputs. The clean archive rebuild deletes generated tables and build outputs before rebuilding offline. None of these verification steps is represented as an editorial decision or independent mathematical peer approval.

### 1.3. 3. R49 blocking and major comments

#### 1.3.1. B1 and M1: One self-contained submission

R51 closed the earlier delivery split. R52 inherits that complete submission and adds the new sources and evidence in an additive directory. The root guide identifies the completed R52 object only after its publication gates pass. The original review and revision branches are not modified. The exact pinned predecessor, both controlling reports and the current delivery identity are separately recorded.

#### 1.3.2. B2 and M2: The identified contribution and neural specificity

The native original-owner distance transform and its exact affine-ReLU realization continue to compute the same continuation, owner and feasible action. They are not two independent empirical methods. The compiler's common saving is not attributed uniquely to a neural architecture.

R52 adds a positive result about what continuation accuracy a Neural Bellman Operator actually needs: an error is relevant through its feasible action contrasts. This goes beyond removal of a datewise constant and includes an explicit non-constant single-ReLU component in the original nonlinear investment model. The coupled-residual argument supplies a constructive certificate rather than assuming the conclusion. The property is appropriately shared by any other representation that supplies the same verified objects. We retain the NBO subject while making this distinction mathematically exact.

#### 1.3.3. B3–B4 and M3–M4: Actual cost and the source of conservatism

The 93 witness-higher, zero witness-lower and three unresolved R49 cost comparisons are unchanged. They remain adverse evidence about those actual policies.

Tighter certificates are not substituted for this ranking. The retained R51 direct study applies the same feasible final-stage improvement to both methods and records each own-incumbent gain and the remaining cross-method difference.

The identity

$$J_W - J_F = (J_W - J_{W+}) + (J_{W+} - J_{F+}) - (J_F - J_{F+})$$

continues to separate removable terminal-action loss from the remaining gap. Earlier-date actions, terminal states, capacity repair, blocked proposals and work remain in the mechanism records. We do not assert that this decomposition identifies every earlier-date source of policy loss.

R52 identifies a separate, precisely removable source of conservatism: a global evaluation-width allowance can reject a beneficial action because of an error whose action contrast is identically zero. The new theorem removes that allowance without changing the policy optimum or weakening the incumbent safety requirement. The exhaustive experiment holds the economic policies fixed and varies only that nuisance. It therefore establishes robustness, not a newly favorable witness-versus-FVI cost ranking. The historical joint cost/work frontiers remain the empirical economic comparison.

#### 1.3.4. B5–B6 and M5: Adaptive conventional methods

The original 72 matched curvature-FVI checkpoints remain explicitly nonadaptive, and their additional overhead is retained. The separately executed graded rule remains genuinely nonuniform and adverse. Neither result is used to dismiss adaptive FVI generally.

The R51 residual/surplus-driven coordinate implementation is retained with its realized nonuniform grids, own-future queries, pilot costs and outcome records. Uniform FVI remains a strong comparator. R52 does not claim that the new certificate ablation is another adaptive-baseline experiment. A broad claim of superiority over sparse grids or other untested adaptive algorithms is not established by these data.

### 1.3.5. *B7 and M6: Dimension and common accuracy*

The retained common-target-five block evaluates both methods in dimensions two, three and four, retaining every failed allocation and refinement, full prefix work, direct costs and peak memory. R52's action-null identity is proved for all declared dimensions, while its exhaustive cell experiment is explicitly two-dimensional. It is not a new dimension-scaling experiment. The tensor state-cover factor and corner-query factor remain in the resource account; neither the new theorem nor this diagnostic removes them.

### 1.3.6. *B8 and M7–M8: Allocation and complete work*

The retained study executes the finite resource planner before each service. Its Bellman-query objective is distinguished from complete construction and verification costs. Failed primary allocations and the separately declared common-target refinements remain visible.

The coupled-residual certificate has additional computational tasks: residual-difference enclosures, coupled integration and a valid feasible-action bound. These tasks are not free. The new study reports the exact mechanism counters, a complete per-service clock through durable cell output, serialization time and peak memory. Its stored records allow all gates to be replayed. These diagnostic clocks are not combined with old construction clocks to manufacture a new economic frontier.

### 1.3.7. *B9 and M9: Information prices and economic interpretation*

The retained finite information technology, precision catalogue, price sensitivity and net-cost selector remain in the current paper. The price parameters are explicit normalized primitives, not estimated technology costs. The certified selector is compared with the available simultaneous cost intervals and their conservative regret account, not an unobserved population optimum.

The new theorem preserves the acquisition restriction. A proposal and its acceptance decision must work on the entire reported cell, and fixed observation precision can leave a positive approximation floor. Removing an action-null error does not remove that information constraint. No new empirical calibration or unsupported welfare interpretation is added.

### 1.3.8. *B10 and M10: Exposition and preservation of the original paper*

We retain the original title and economic subject. The active article develops a self-contained constructive policy-evaluation, feasible-improvement, resource and evidence chain. Complete development editions preserve the original broad theory and applications, with their assumptions and historical descriptions distinguished from the currently executed experiment. Every predecessor label is checked. The new main and supplementary sections are added to both complete editions rather than replacing earlier applications with a new topic.

Linear action-kernel contrast results are stated for expected discounted costs. Earlier nonlinear-recursion safety results remain available under their own hypotheses. Preservation is not a claim that every historical application has received the new numerical experiment, and the new theorem is not silently extrapolated to nonlinear preference recursions.

### 1.4. *4. Interpretation of the completed revision*

R52 supplies a stronger proved accuracy account, an executable residual-based certificate, a nonconstant neural action-null realization in the original model, and a complete deterministic robustness test with ordinary source and evidence. These additions directly address the distinction between critic error, safe policy improvement and actual economic cost. They do not reverse the inherited unfavorable method comparisons. Ten new exact tests supplement the 44 inherited tests; the executed test totals and publication outcomes are recorded by the build rather than assumed in advance.

The remaining empirical scope is stated precisely: full-horizon investment improvement, high-dimensional non-tensor performance, comprehensive adaptive-method superiority and empirically estimated sensor prices are not established by the new terminal ablation. These are not substituted for the original theorem targets or used to abandon the NBO paper. The revision provides its mathematical statements and reproducible evidence for the referee to evaluate on their merits.